

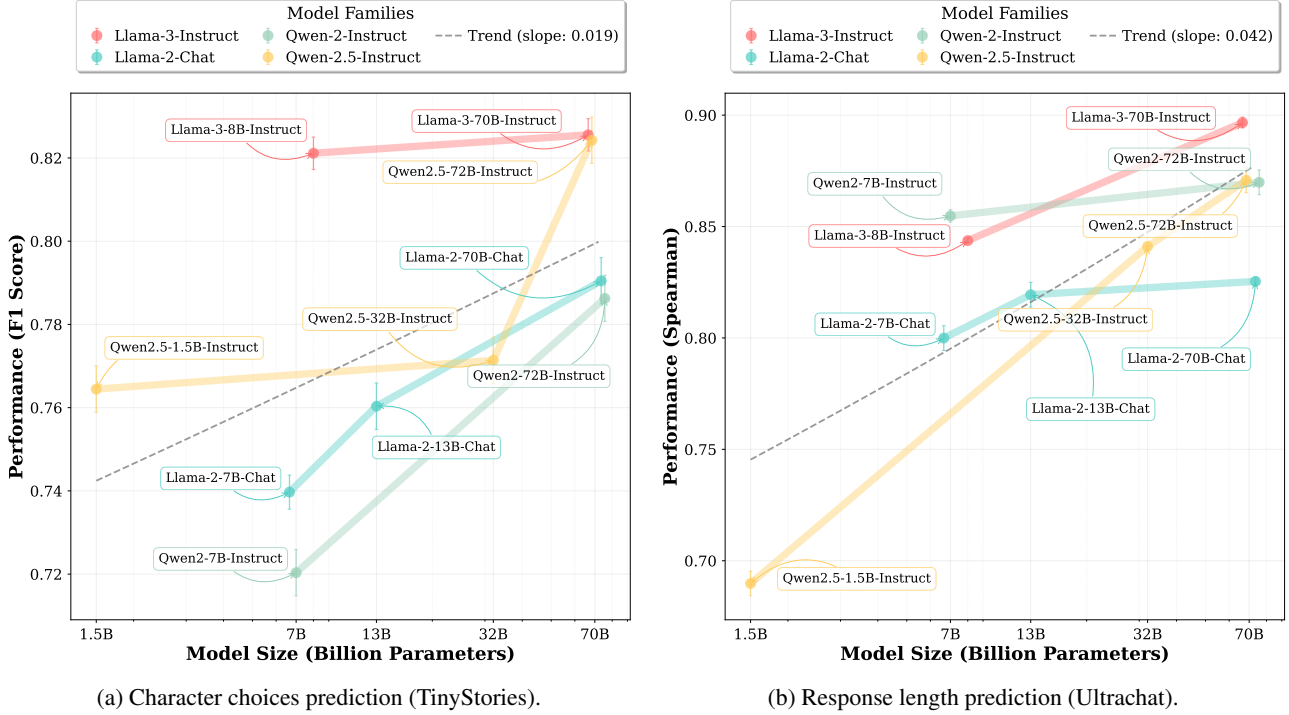


Figure 6. Scaling effects on planning capabilities. Evaluated across four model families (Llama-2-chat, Llama-3-Instruct, Qwen-2-Instruct, Qwen-2.5-Instruct; 1.5B–72B) using UltraChat and TinyStories, structure and content attributes show family-specific scaling: larger models within each family improve planning.

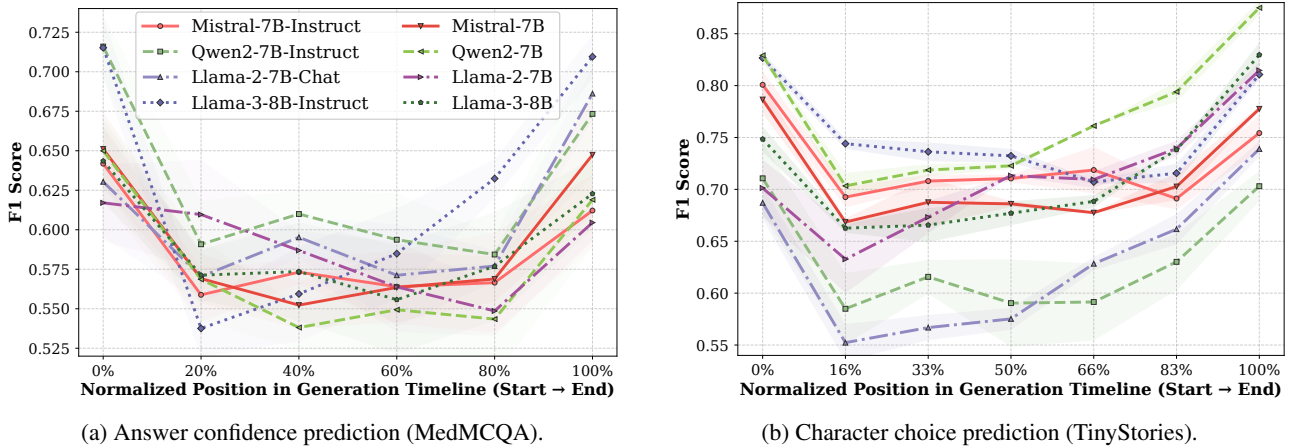


Figure 7. Planning dynamics during generation. Probing at equidistant positions (answer confidence, character choice) shows three-phase patterns: high accuracy in early segments (global planning intent), mid-segment decline (local token focus), and late-stage recovery (contextualized refinement). This suggests models first outline global attributes, then refine locally, before finalizing coherent plans.

optimal configurations and evaluate models on UltraChat and TinyStories datasets, focusing on structure and content attributes. We exclude base models as smaller models have short context which limit few-shot prompts, while the same prompts fail to effectively prompt larger base models to follow instructions. We omit the behavior attribute type as larger models tend to give correct answers consistently, making it difficult to obtain balanced data for analysis.

Fig. 6 exhibit two key insights: **(1)** within each model fam-

ily, larger models demonstrate stronger planning capabilities, and **(2)** this scaling pattern does not generalize across different model families, suggesting that other factors like architectural differences also influence planning behavior.

5.2. Evolution of Planning Representations During Response Generation

We analyze how planning features evolve during generation by probing at different positions in the response sequence.

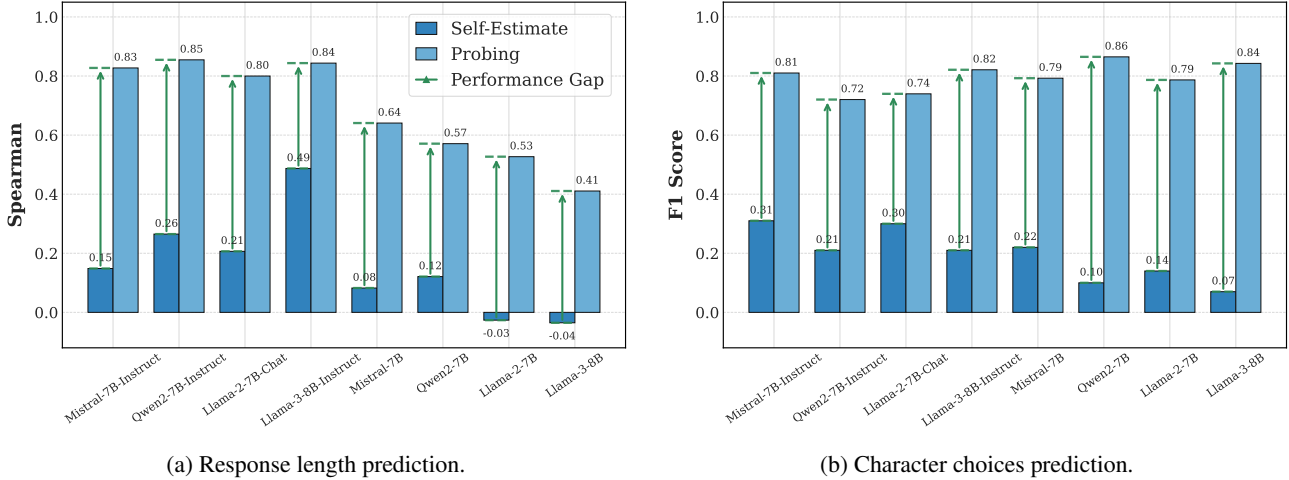


Figure 8. Gap between probed and verbalized results. Both tasks reveal a systematic gap between verbalized self-estimates (*Self-Estimate*) and probe-based predictions (*Probing*). Base models exhibit near or worse-than-random accuracy in self-estimation, while fine-tuned models achieve only marginal gains. This gap demonstrates that models encode richer planning information in hidden representations than they can explicitly access during generation, revealing a divide between implicit planning and explicit self-awareness.

For each response, we collect activations from the first token up to the token before attribute-revealing keywords (e.g., animal words in story character selection tasks) or throughout the entire sequence for tasks requiring external ground-truth labels (e.g., answer confidence tasks). We divide these positions into equal segments and apply probes previously trained with in-dataset settings at each division point. We conduct experiments on two datasets: TinyStories for character choice prediction and MedMCQA for answer confidence prediction. Fig. 7 reveal a distinctive pattern: **probing accuracy is high initially, decreases in the middle segments, and rises again toward the end**. This pattern suggests a three-phase planning process: (1) initial phase with strong planning that provides an overview of the intended response; (2) middle phase with weaker planning, characterized by more local, token-by-token generation; (3) final phase with increased planning clarity as accumulated context makes the target attributes more apparent.

5.3. Gap Between Probed and Verbalized Results

We investigate whether LLMs can self-estimate their response attributes when explicitly prompted, and compare these verbalized results—self-predictions obtained via direct prompting—against probe-based predictions. Experiments focus on two tasks: response length prediction (Ultrachat) and character choice (TinyStories). For verbalized predictions, we prompt LLMs in separate runs to self-estimate attributes (e.g., “Estimate your answer length in tokens using [TOKENS] number [/TOKENS]” for tuned models, or via few-shot examples with pre-calculated lengths for base models). Self-estimation accuracy is evaluated by comparing these outputs against actual greedy-decoded response

attributes.

Fig. 8 reveals a systematic gap: base models exhibit near or worse-than-random accuracy in self-estimation, while fine-tuned models improve only marginally, remaining far inferior to probe-based methods. This suggests **models encode more planning information in hidden representations than they can explicitly access during generation**, highlighting a divide between implicit planning and explicit self-awareness.

6. Discussion

6.1. Emergent Response Planning under Sampling

In this study, we consistently use greedy decoding to derive deterministic probing labels $\hat{g}_i = g(\mathbf{y}_i)$ for representations $\mathcal{H}_i = \{\mathbf{H}_{\mathbf{x}_i}^l\}_{l=1}^L$. But when generalizing to sampling settings, while greedy decoding simplifies sampling approximation by reflecting the LLM’s most probable output, this approach may not fully capture sampling nuances. We propose two potential ways for improvement:

Averaging: Replace greedy labels with attribute averages over multiple sampling trials (e.g., 10 samples) to approximate expected sampling behavior.

Distributional probing: Train probes to predict label distributions instead of single values, capturing uncertainty inherent to sampling. While greedy decoding reflects the LLM’s most probable output (approximating sampling averages), distribution-aware probing remains an open challenge, which we leave for future work.

6.2. Defining Planning and Addressing Spurious Correlations

A crucial consideration in defining and measuring “planning” is the potential for spurious correlations, particularly **“first-token shortcuts.”** An example illustrates this: if a model is prompted in either French or English to provide a ‘yes/no’ answer in the same language, a probe analyzing the first token’s activations might predict the final answer (“oui/non” vs. “yes/no”) simply by detecting the language of the initial token. While this shows a correlation, it doesn’t necessarily prove long-range planning in the sense of anticipating specific future content beyond what’s implied by immediate context like language choice.

Our study addresses this by (1) defining planning as the encoding of long-term attributes independently of the immediate next token, and (2) designing prompts to actively block these shortcuts. We define planning such that the hidden representations at the first token should encode both next-token information and long-term attributes, ensuring these two information types are independent – the long-term attribute shouldn’t be directly inferable from the very next token. To implement this in our experiments, we used prompt engineering. For instance, in multiple-choice tasks, we instructed models to first provide an analysis before stating their final answer. This ensures that the initial tokens (the analysis) do not inherently reveal the target attribute (the final choice), thereby helping to isolate genuine planning signals from simpler, shortcut correlations. This methodological approach is vital for ensuring that probes detect true emergent planning rather than just correlated input features.

6.3. Potential Applications of Emergent Response Planning in LLMs.

Our findings on LLMs’ emergent response planning suggest several practical applications: (1) **Pre-generation resource allocation optimization:** Probing pre-generation representations allows proactive allocation of computational resources based on anticipated response complexity and length, enhancing inference efficiency during dynamic workloads. (2) **Early-error detection:** Early detection of behavioral attributes like low confidence could enable corrective interventions (e.g., retrieval-augmented refinement) before errors propagate. Predictive awareness of content attributes (e.g., key entities or argument trajectories) might enable real-time compliance checks with safety guidelines or domain constraints. (3) **Novel user interaction paradigms:** Predicting reasoning complexity could guide task decomposition for multi-step problems, while predicting response characteristics could improve progress indicators in interactive settings. These possibilities highlight the need for robust probing methods in deployed LLMs.

6.4. Future Research Directions

Several key research directions emerge from our findings: (1) **Causal mechanisms of planning:** Research could investigate whether and how planning representations directly influence token generation (e.g., via causal intervention experiments). Establishing causality is crucial for reliably leveraging these representations and understanding LLM decision-making. (2) **Leverage planning for generation control:** Future work might explore methods to detect and utilize pre-generation attribute predictions (e.g., key content points) for real-time steering. This could enable more efficient and precise control than post-hoc correction, potentially reducing computational waste and errors by allowing early adjustments based on predicted response properties. (3) **Planning in multimodal contexts:** Exploration of whether similar planning phenomena emerge in multimodal (e.g., image+text) LLMs could be valuable. Such research may provide insights into the universality of emergent planning and how modality impacts the development of these cognitive capabilities. (4) **Planning-aware training:** Developing objectives that explicitly reward alignment between early-plan encodings and final outputs (e.g., via consistency losses) represents another avenue. This may enhance coherence in complex tasks by grounding generation in initial intent.

7. Conclusion

In conclusion, our work reveals that LLMs have emergent response planning capabilities, with prompt representations encoding global attributes of future outputs across structure, content, and behavior attributes. These findings challenge the conventional view of LLMs as purely local predictors and provide new insights into their internal mechanisms. Though we do not focus on interpretability mechanisms to explain the causal relationship of emergent response planning, our findings open promising directions for enhancing model control and transparency, potentially enabling more effective methods for guiding and predicting model outputs before generation begins.

Impact Statement

Our findings on LLM emergent planning raise specific considerations for model deployment. While these capabilities could enhance system reliability through better resource allocation and early warning mechanisms, they also present concerns when handling sensitive data, as these probing methods reveal aspects of the model’s internal thinking or decision-making process. We encourage careful evaluation of these trade-offs when implementing probing-based monitoring systems, particularly in applications involving sensitive information.